

# *An Introduction to 3D Printing*

*Part 2: Practical Application*

*Examples, Critical Thinking, and Design*

# Outline

## Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

## Part 2: Practical Application

Examples

Critical Thinking

Design

## Part 3: Science

Vibrations and Waves

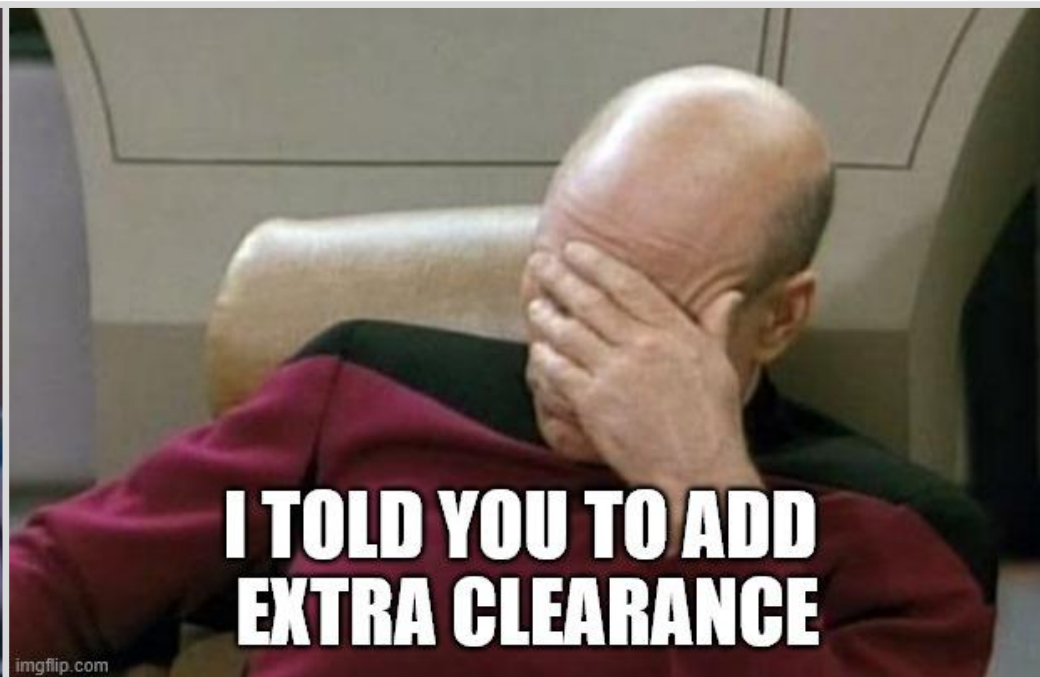
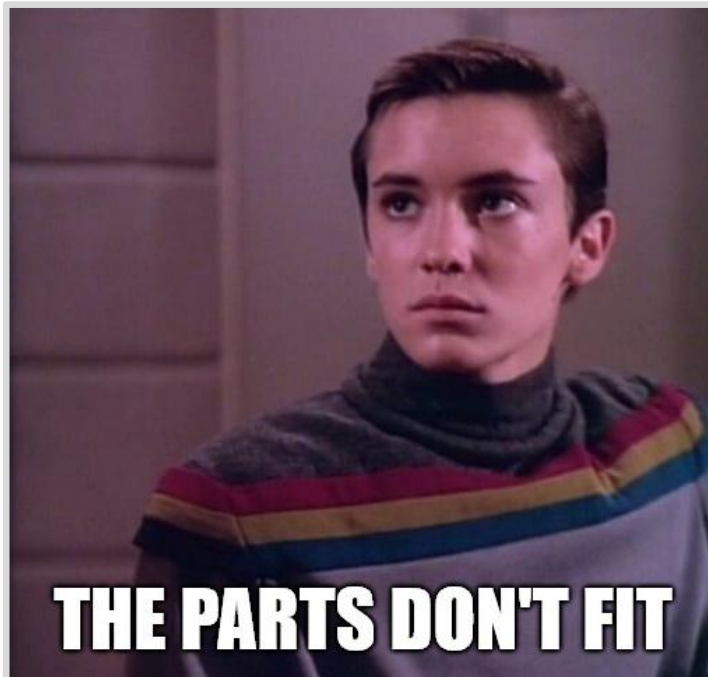
Linear Motion

Thermal Expansion

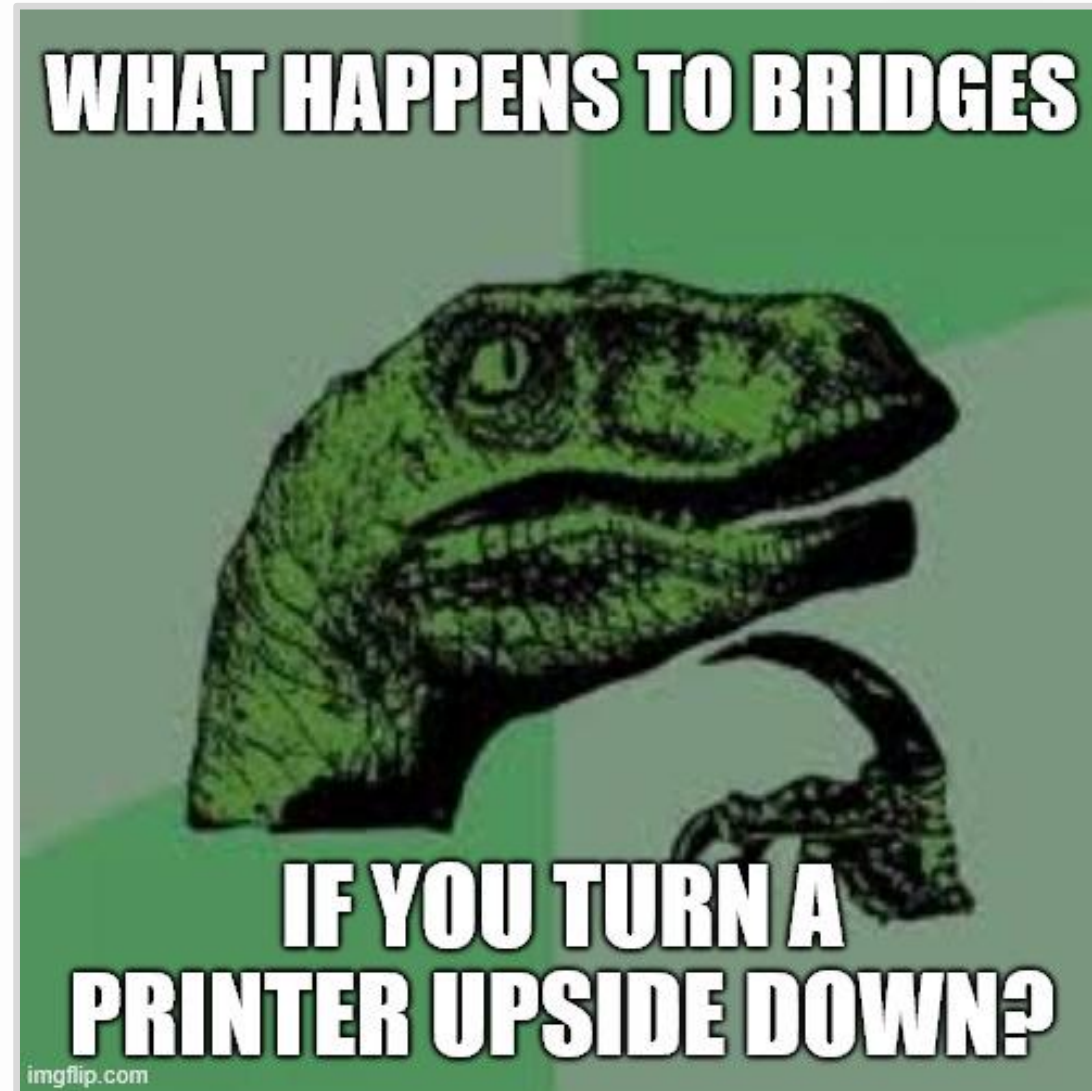
# Pop Quiz

How do you help printed parts fit together?

- You sometimes need scale factor.
- You always need constant offset.



# Anti-Gravity?



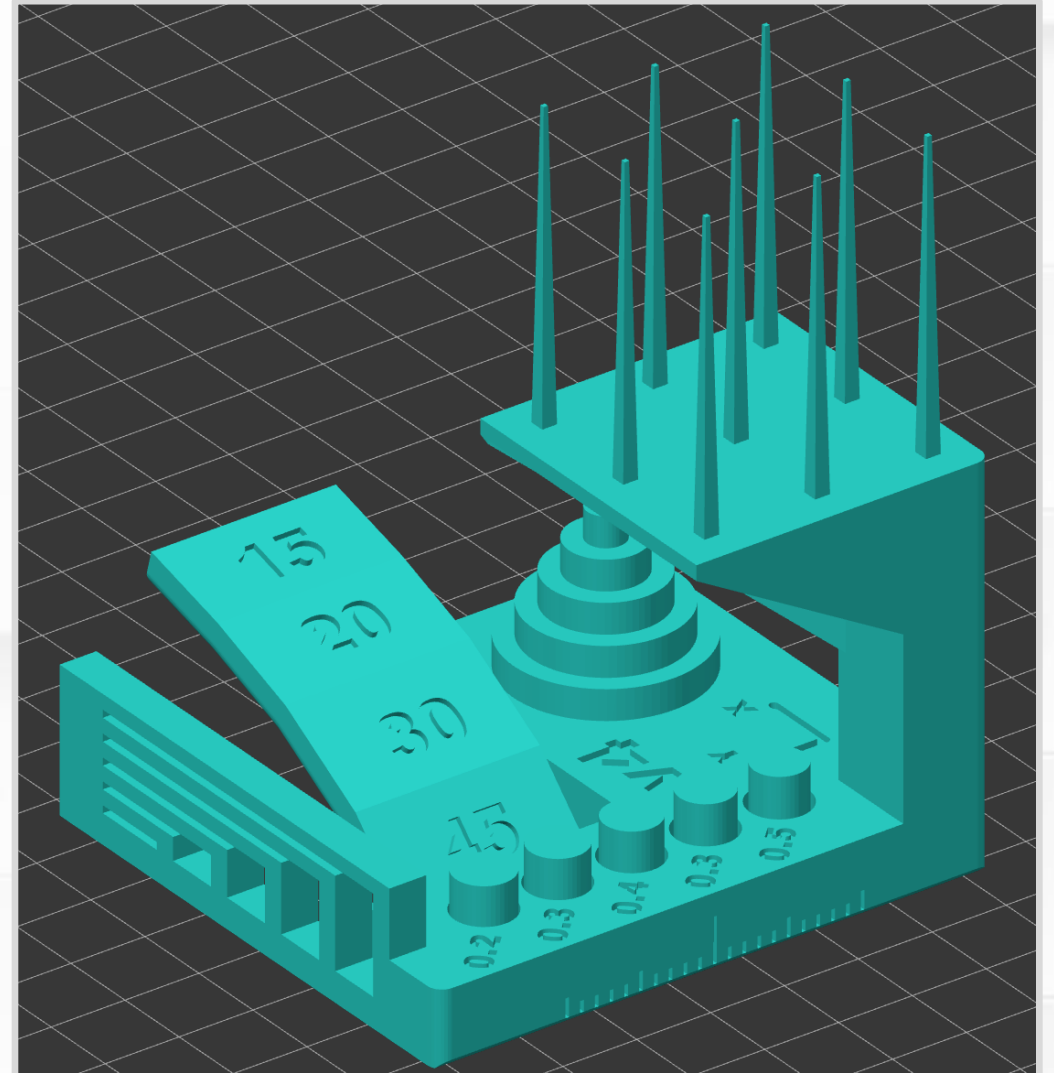
# Anti-Gravity

- Positron GitHub.
  - Prints upside down.
  - Folds up and fits in a small box.
- Design team and users report the same print quality. Why?
- Extrusions hold their shape when you press them onto a solid surface.



# Stress Testing

- Autodesk Test Print. One of many test prints.
- Difficult shapes that highlight capabilities and limitations of your printer + filament.
- What are these shapes testing?
- Full breakdown [here](#).
- Bridges, overhangs, print-in-place tolerance, dimensional accuracy, ...



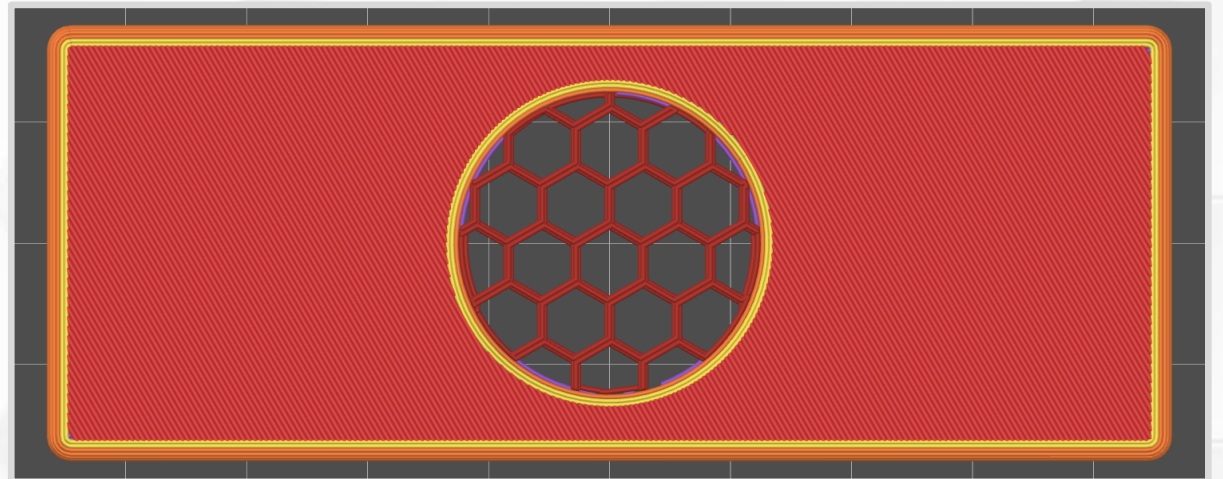
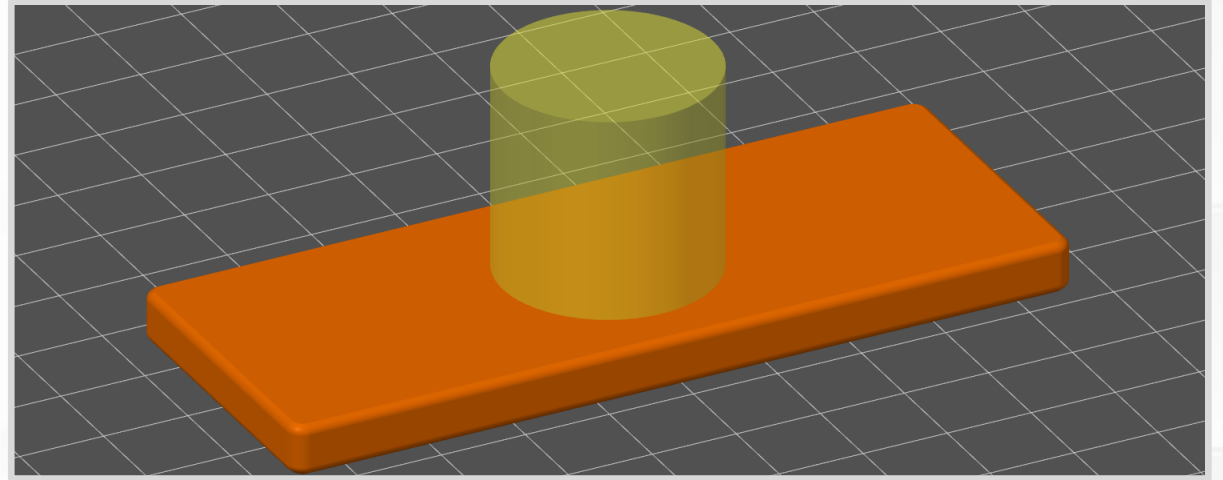
# Corners

- Rounding corners with fillets is a good idea. Why?
- Stresses concentrate in corners.
  - Adding fillets improves strength.
  - Fillets on bottom surfaces reduce first layer warping.
- Print faster with more accurate movements. Why?
  - Toolhead is like a racecar going around a track.
  - Smoothing changes of direction means:  
Less slowdown.  
Less vibrations.

# Advanced Slicing: Modifiers

Let's make a mesh grill with PrusaSlicer.

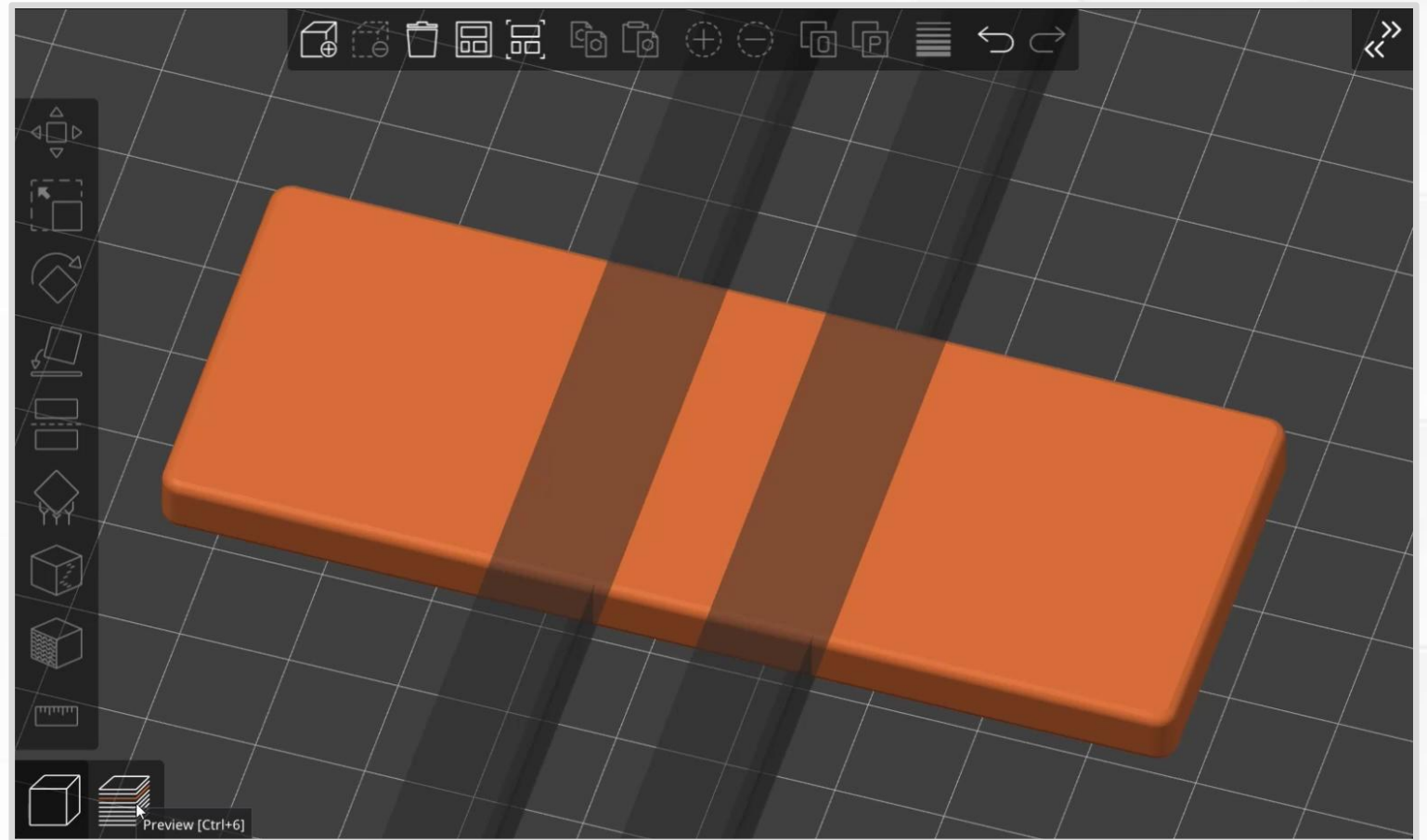
- Load rectangle STL file (orange).
- Add basic shape (yellow) and change it to a modifier.
- Assign zero top and bottom layers.
- Set an infill pattern.
  - Hexagon makes a nice mesh.
- Change infill percent to alter the mesh size.



# Advanced Slicing: Negative Volume

Now let's make a print-in-place hinge with PrusaSlicer.

- Start with a flat object.
- Download [Flexify template](#) that matches part height.
- Templates define negative space for PIP hinges.

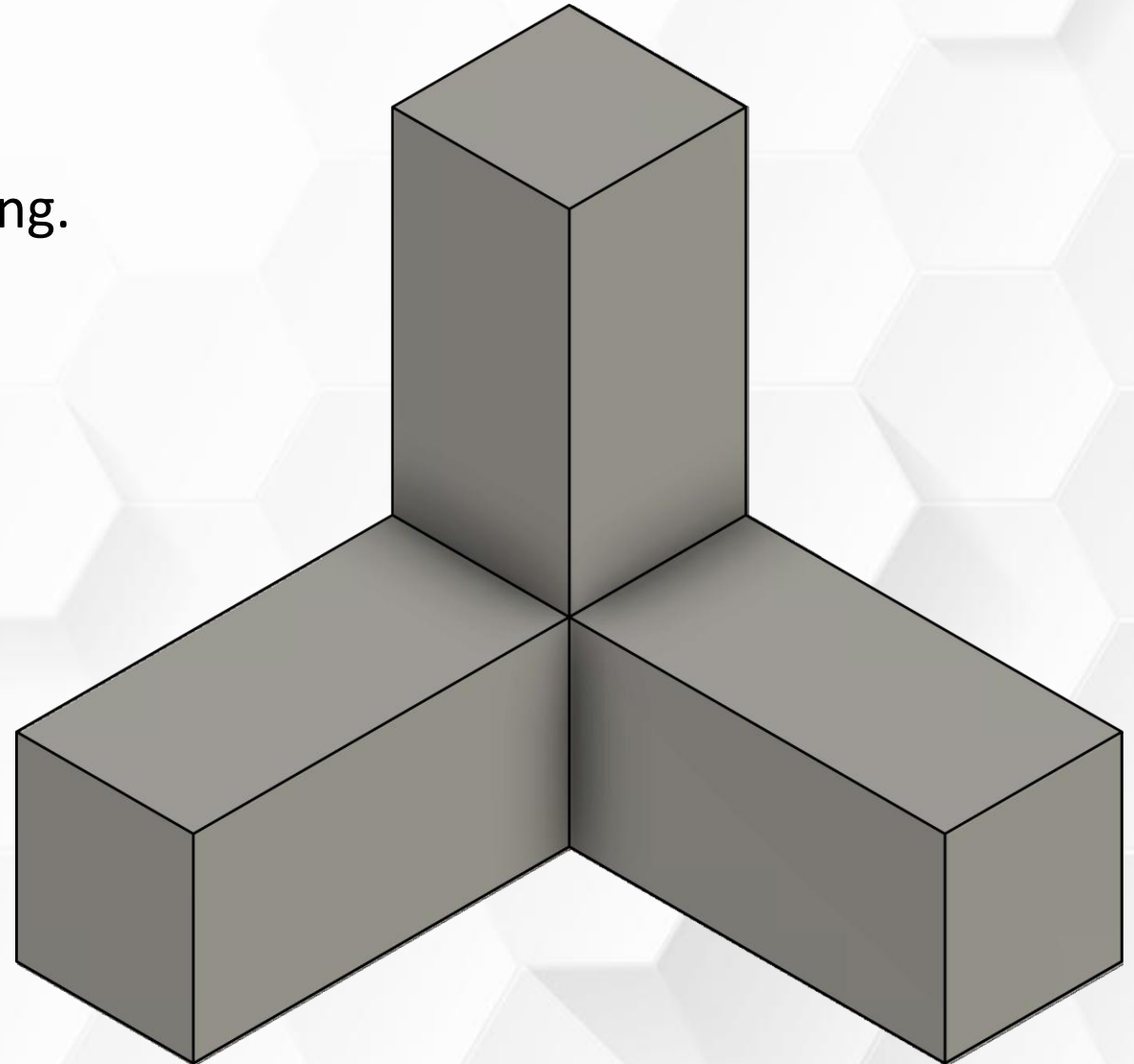


(Animation, [FlyingGyroscope GitHub](#))

# Corner Bracket

Let's improve this 3-way corner bracket.

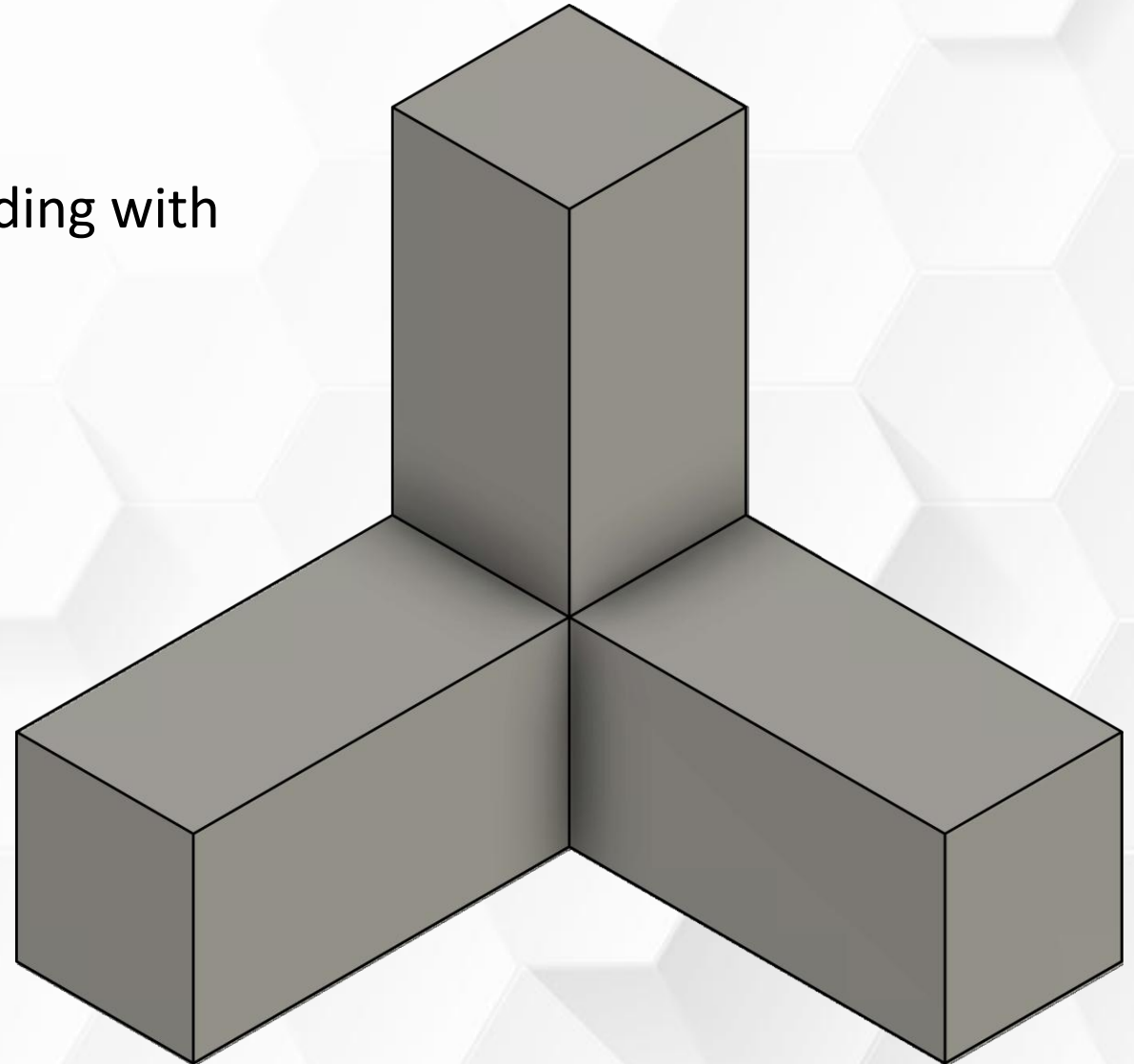
- Top section is weaker because of layer stacking.
- How can we make it stronger?
  - Change slicer settings?
  - Modify or reimagine the design?



# Slicer Settings

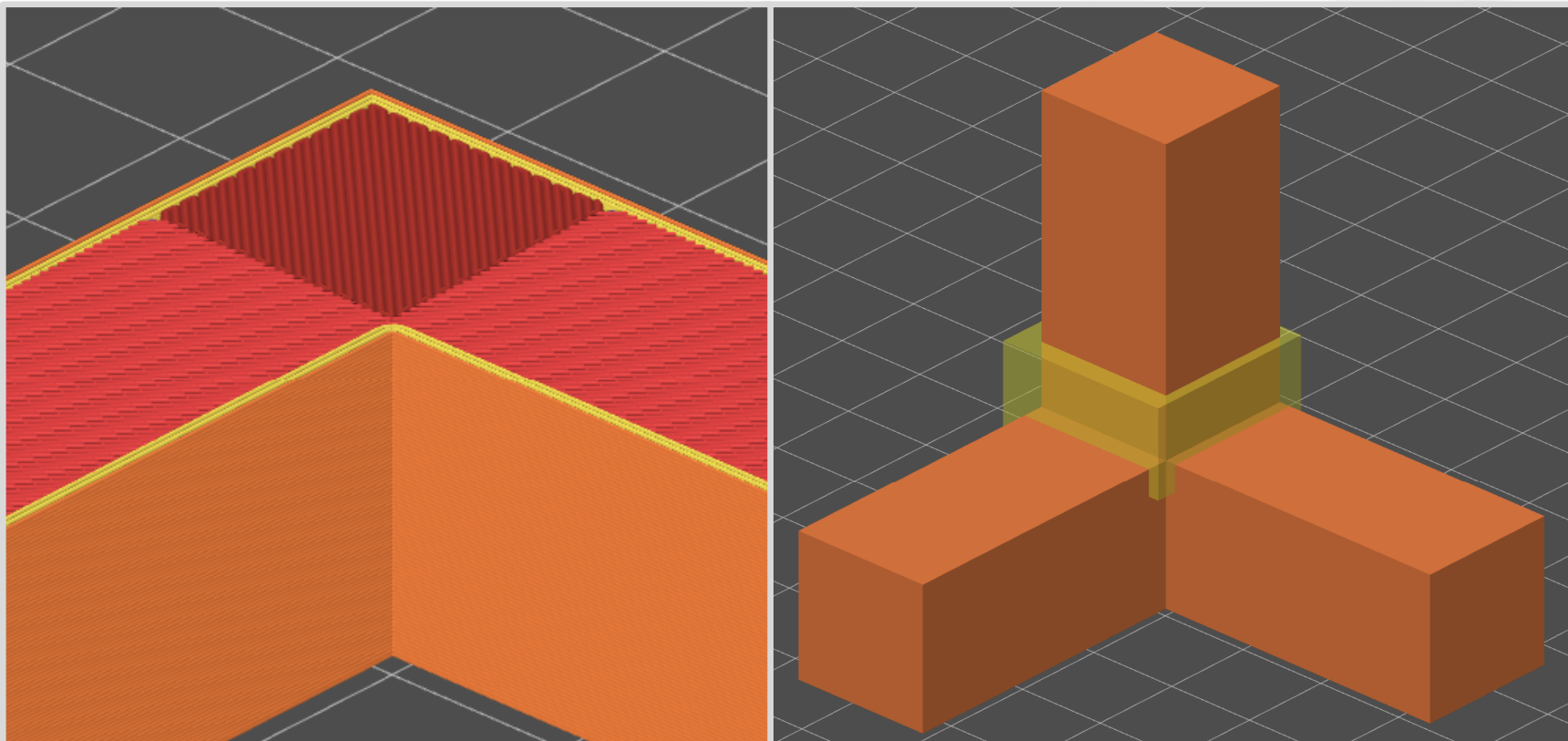
Let's start with slicer settings.

- How can we improve strength and layer bonding with slicer settings?
- Higher temperature.
- Slower speed.
- Less cooling.
- More perimeters > Higher infill density.
- Solid infill.
- Intentionally over-extrude. Eliminates any gaps between extrusions.



# Slicer Settings

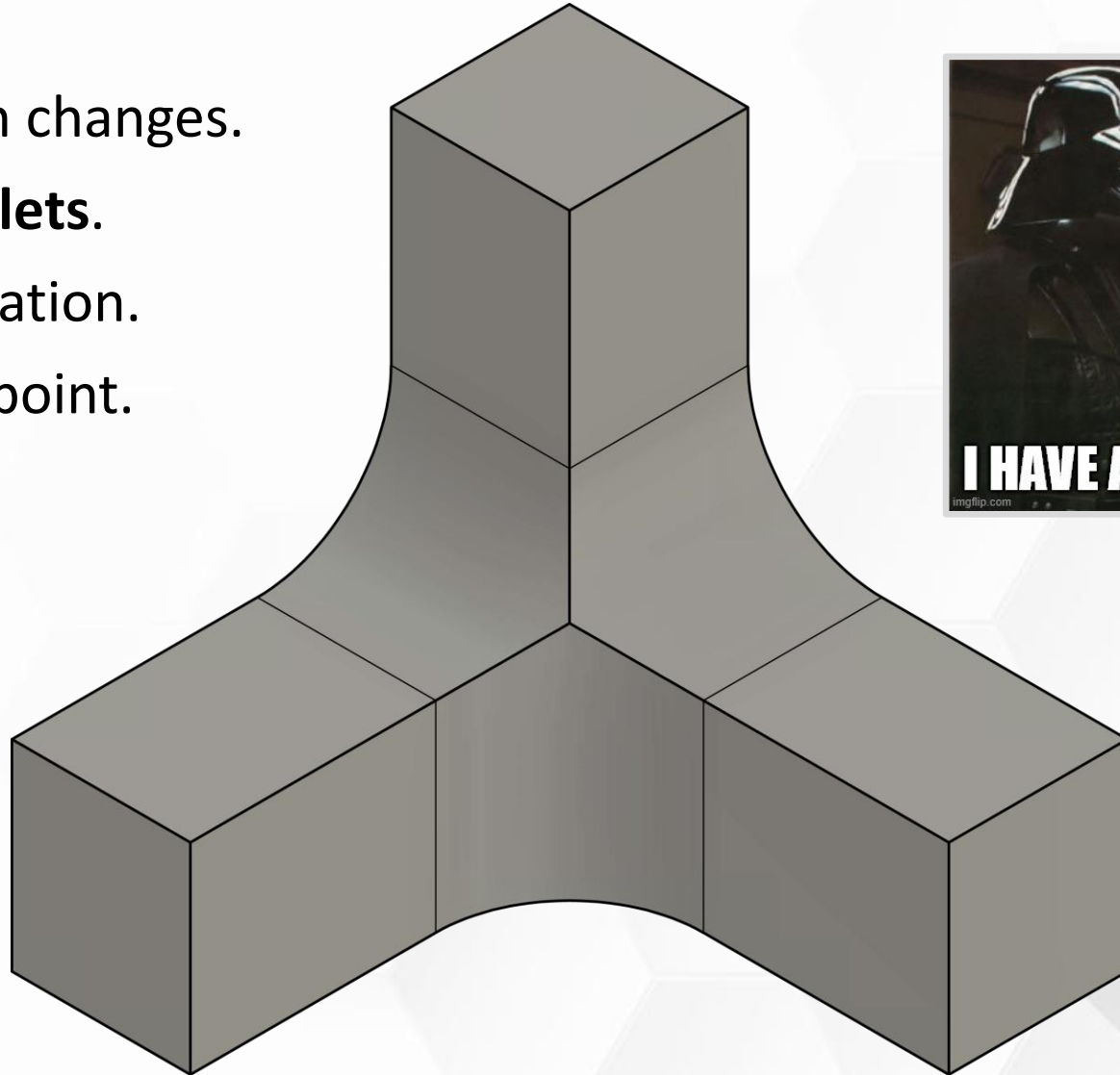
- Use modifier on just the weakest spot. (Yellow box on the right.)
- Assign 100% infill. Maximum surface area for layer bonding.



# Corner Bracket Remix

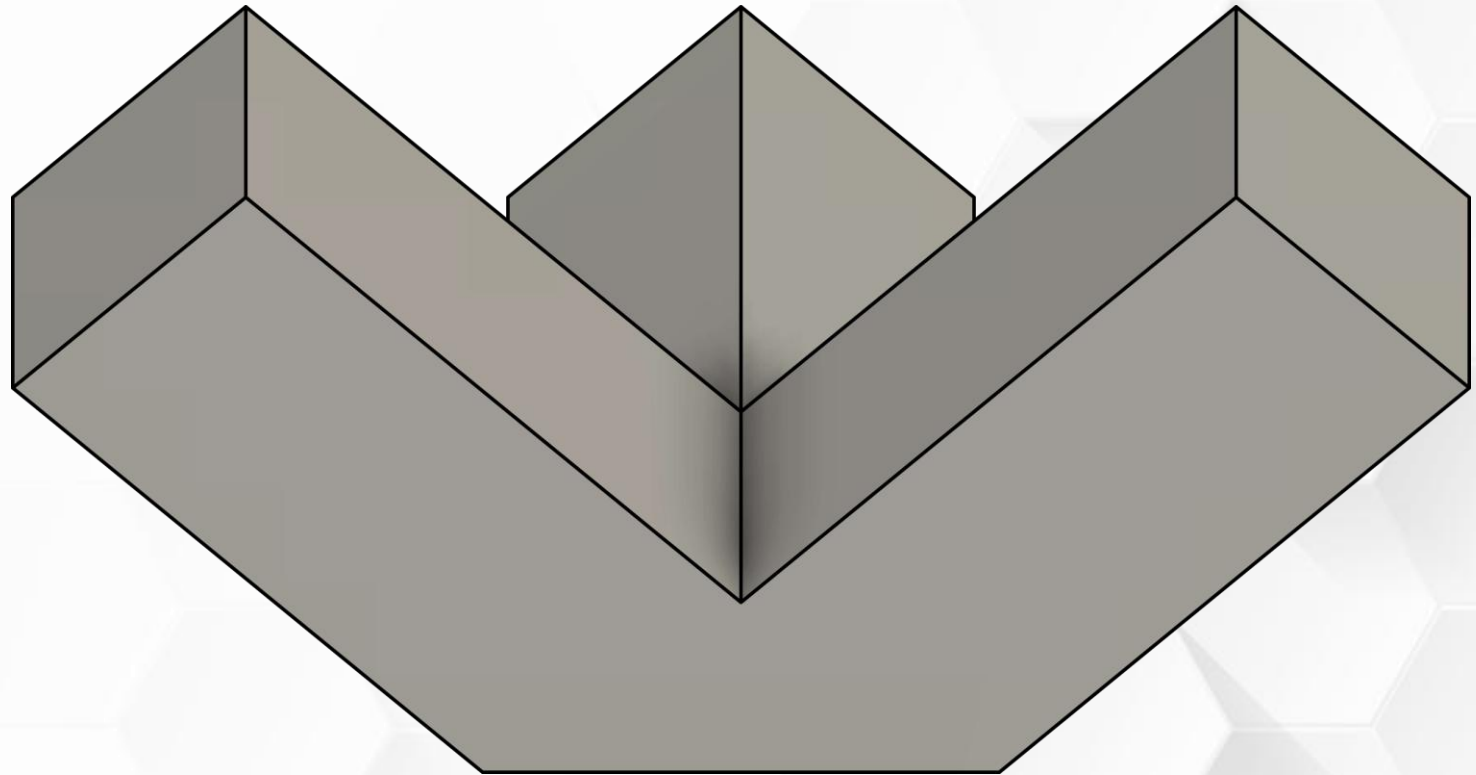
Now let's consider design changes.

- Reinforce joints with **fillets**.
- Reduce stress concentration.
- Reinforce the weakest point.
- Simple and effective.



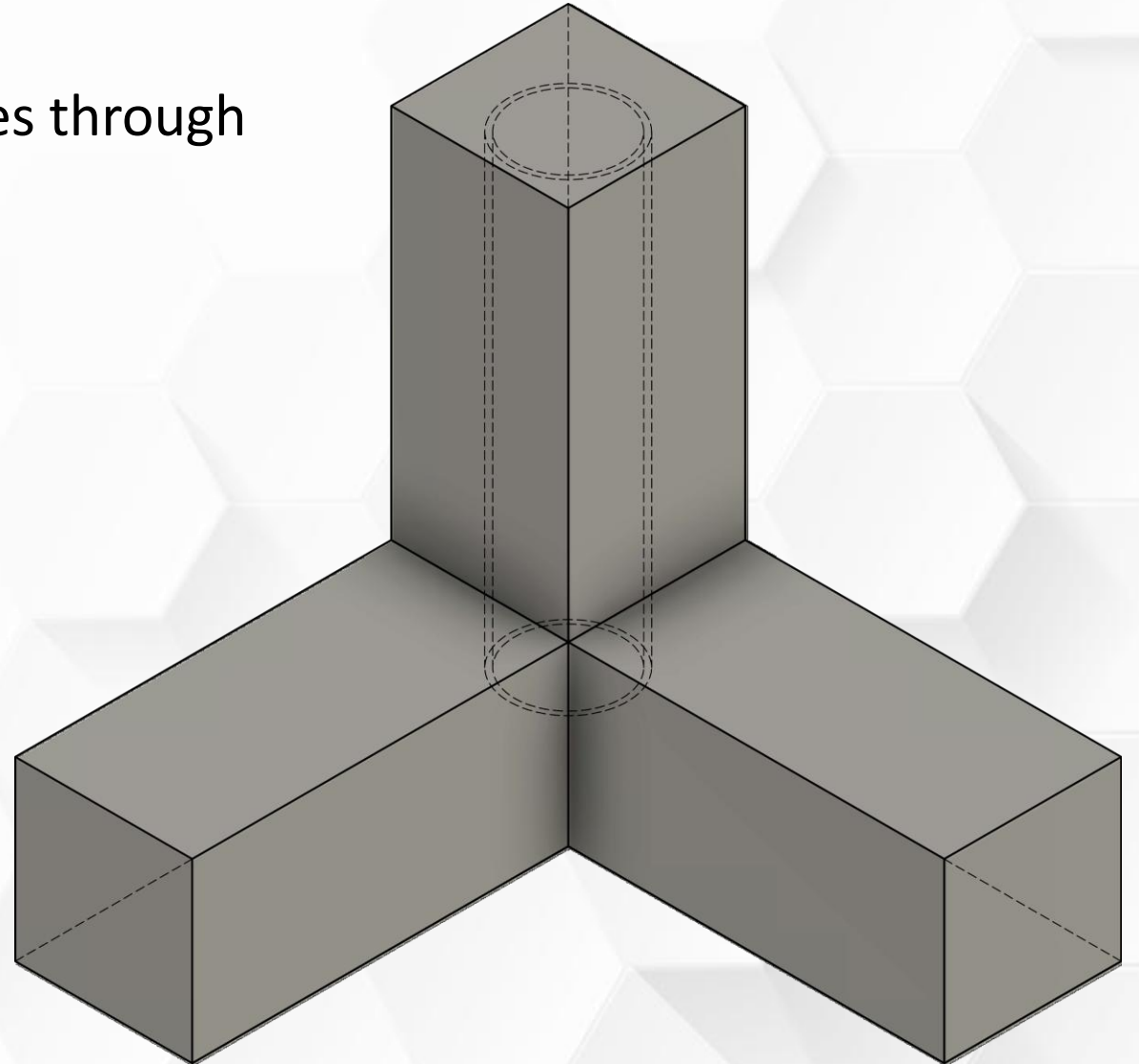
# Corner Bracket Remix

- Angled print orientation.
- Trim the corner.
- Adds overhangs.
- Smaller first layer.
- Each section has equal strength.



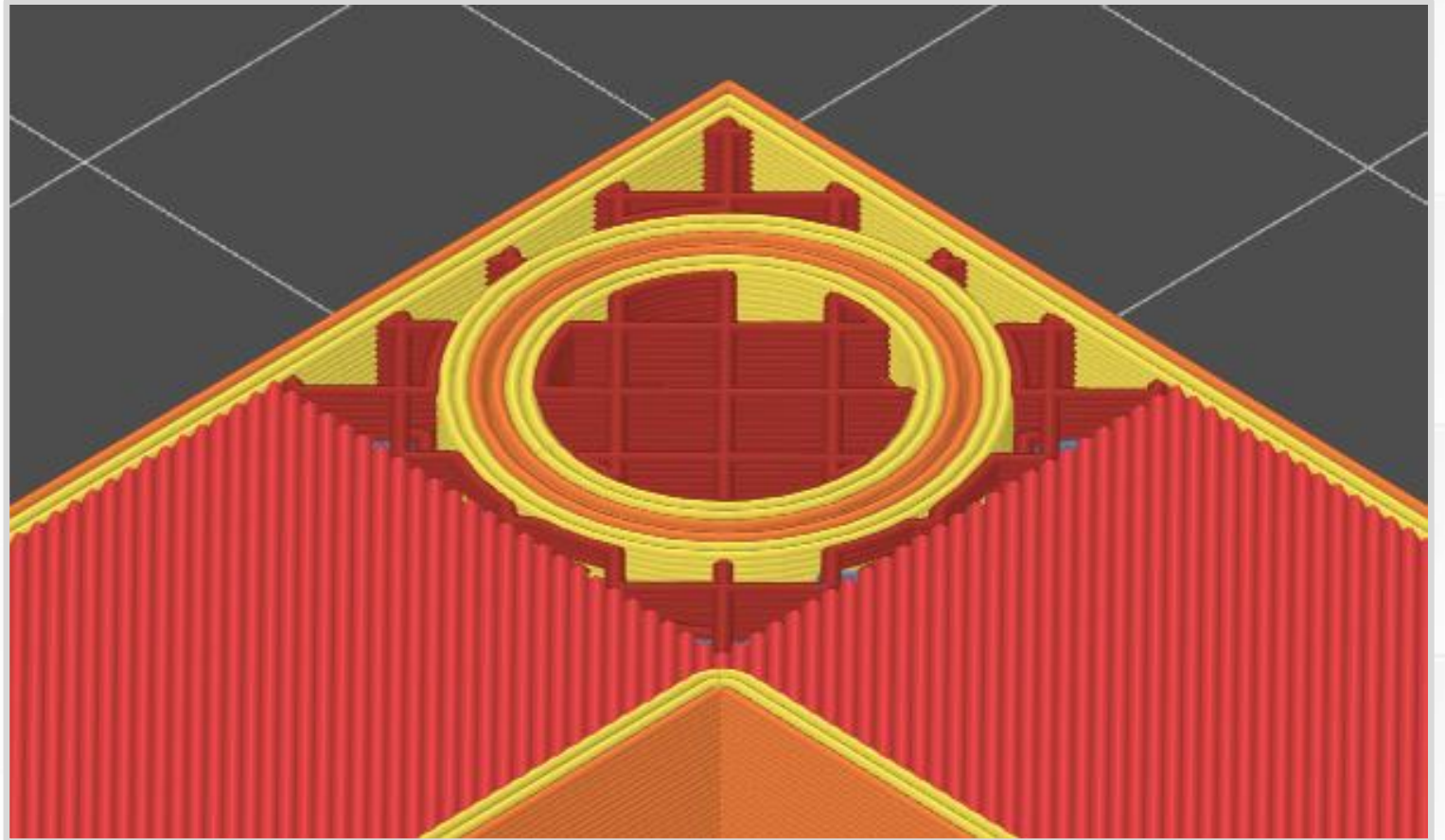
# Corner Bracket Remix

- Cut a small slit (around 0.1 mm) that goes through upper and lower sections.
- Why?
- *Hint:* What will the slicer do?



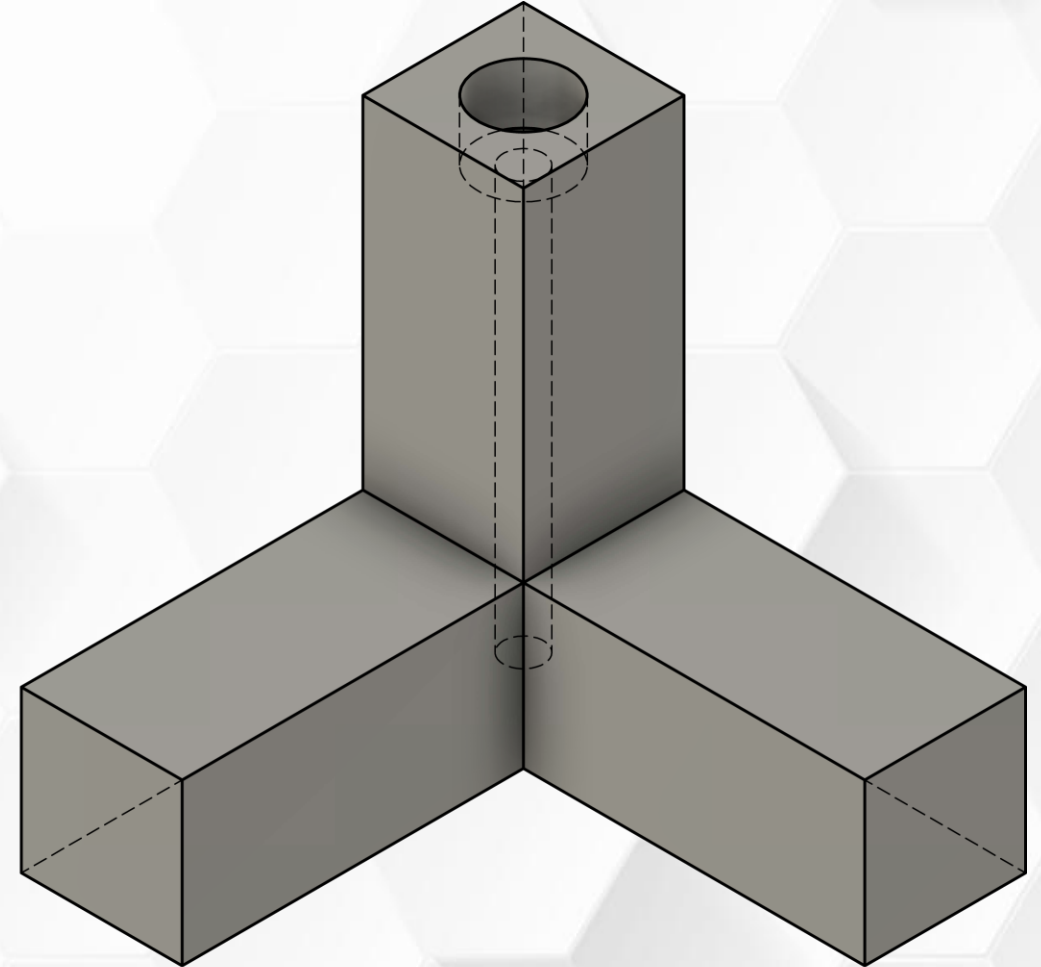
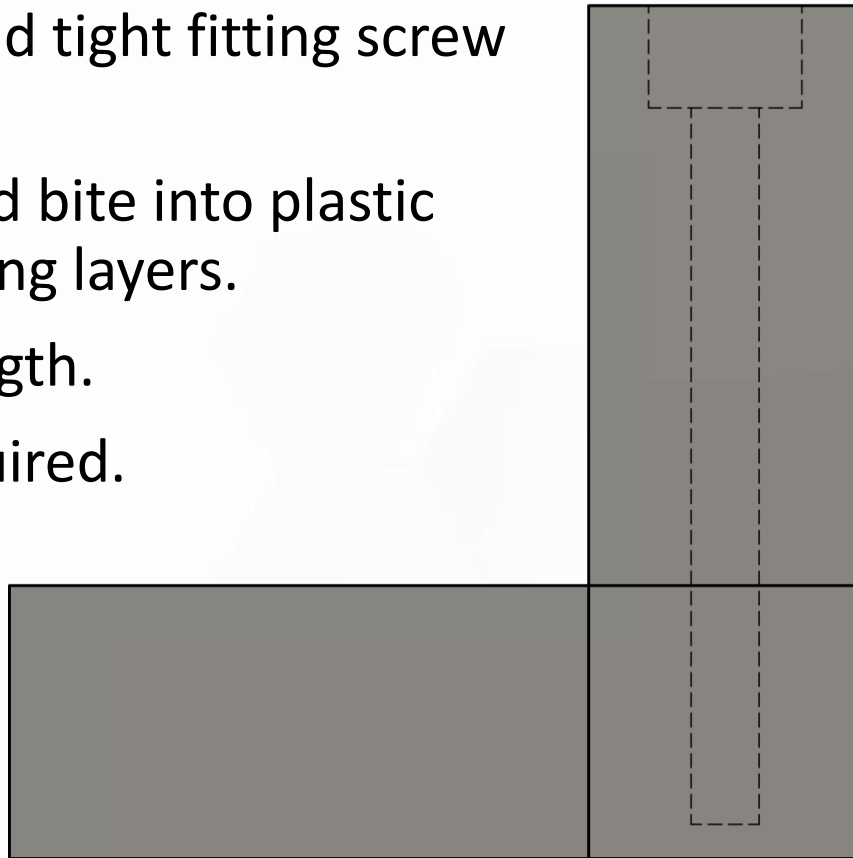
# Corner Bracket Remix

- Tricks slicer into adding extra perimeters.
- More area for layer bonding.
- Less effective than 100% infill modifier from earlier.



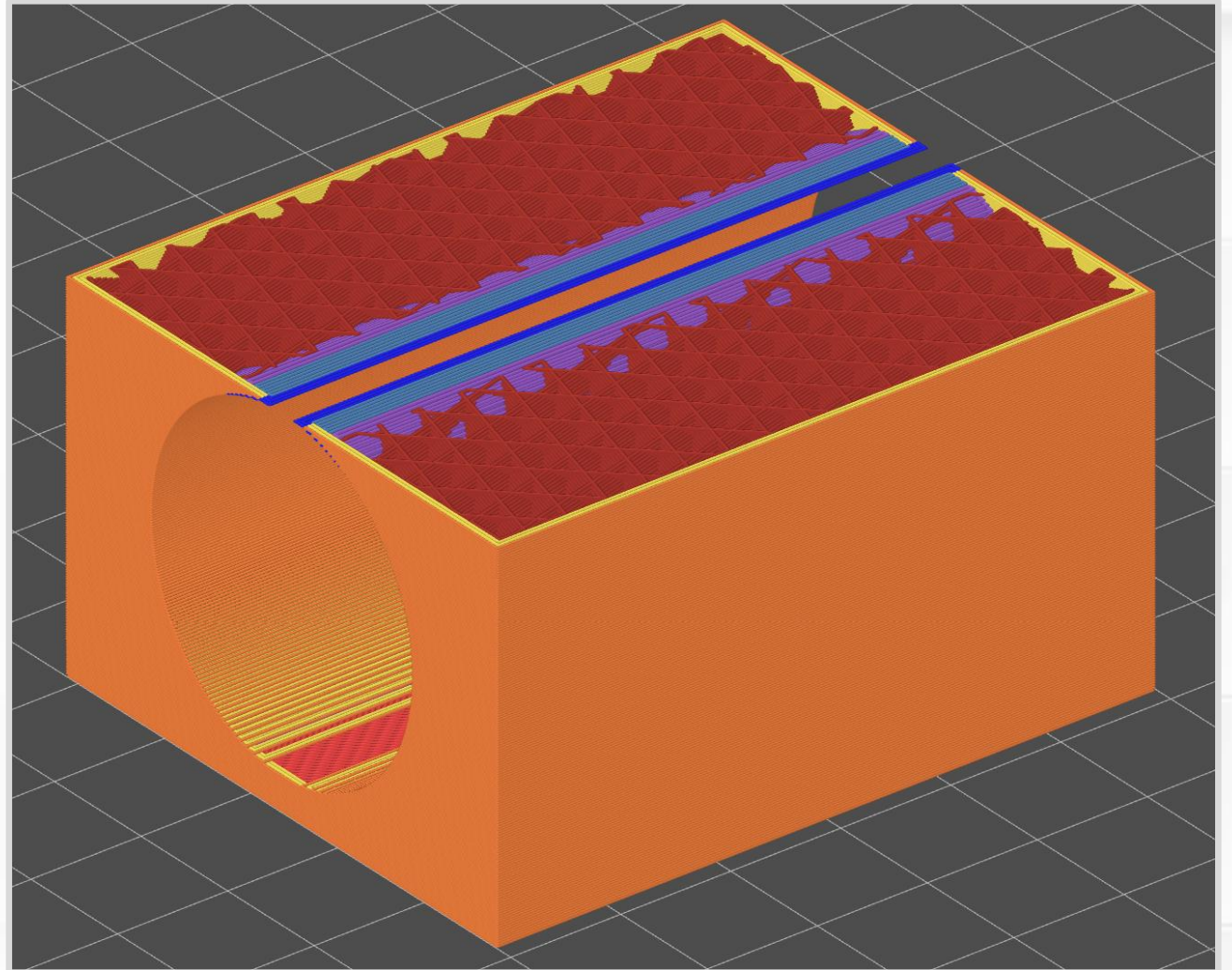
# Corner Bracket Remix

- Borrow strength from something else.
- Cut a hole. Add tight fitting screw after printing.
- Threads should bite into plastic without cracking layers.
- Superior strength.
- Assembly required.



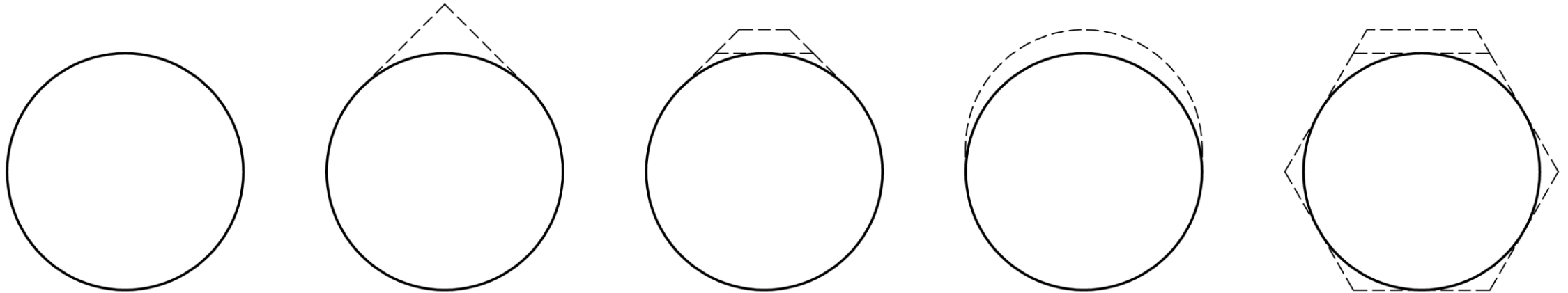
# Horizontal Holes

- Horizontal holes have steep overhangs at the top.
- Hole is undersized if the extrusions collapse.
- Assume the hole must remain horizontal.
- How can you alter the design to fix the problem?



# Horizontal Holes

- Make the top profile more printer-friendly.
- Add clearance at the top.



# Bridges

- What can you do to improve bridges? (Besides supports.)
- *Hint:* How can you help extrusions lock in place?
  - Lower filament temperature.
  - Slower printing speed.
  - Increase cooling.
- Reimagine the object as multiple pieces.
  - Each piece is free from supports and bridges.
  - Assembly and clearances required.
- Can you reimagine the bridge?

# Sequential Bridges

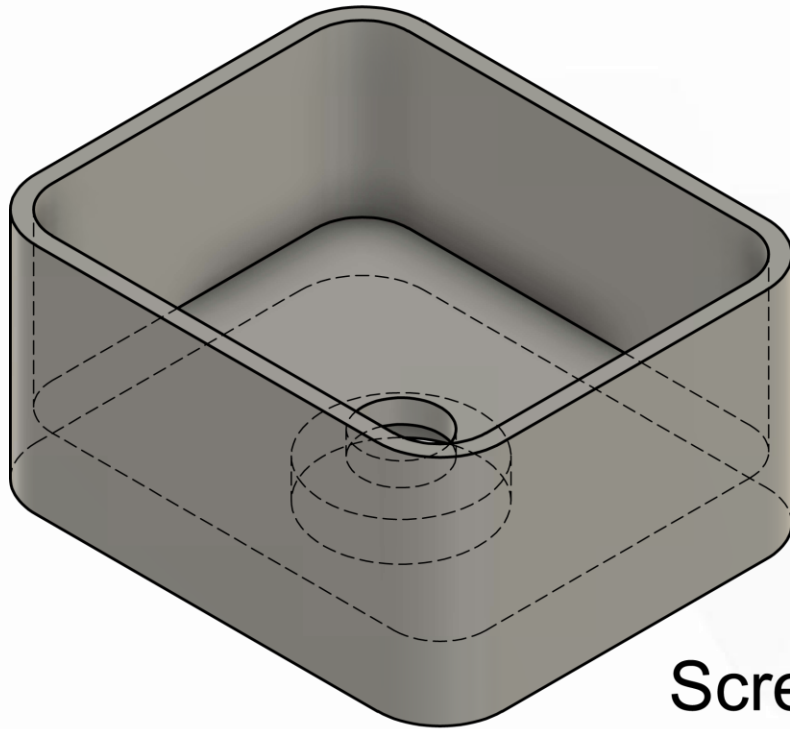
Let's combine small bridges to create a larger, multi-step bridge.



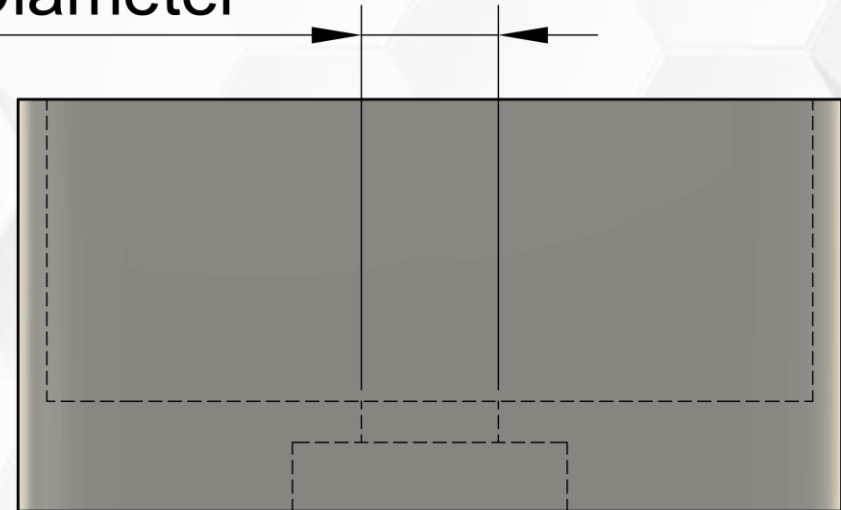
# Sequential Bridges

Consider a storage box with a countersunk screw on the bottom.

- Can you see the hidden problem?



Screw Diameter



Screw Head Diameter



WARNING:

Detected print stability issues:

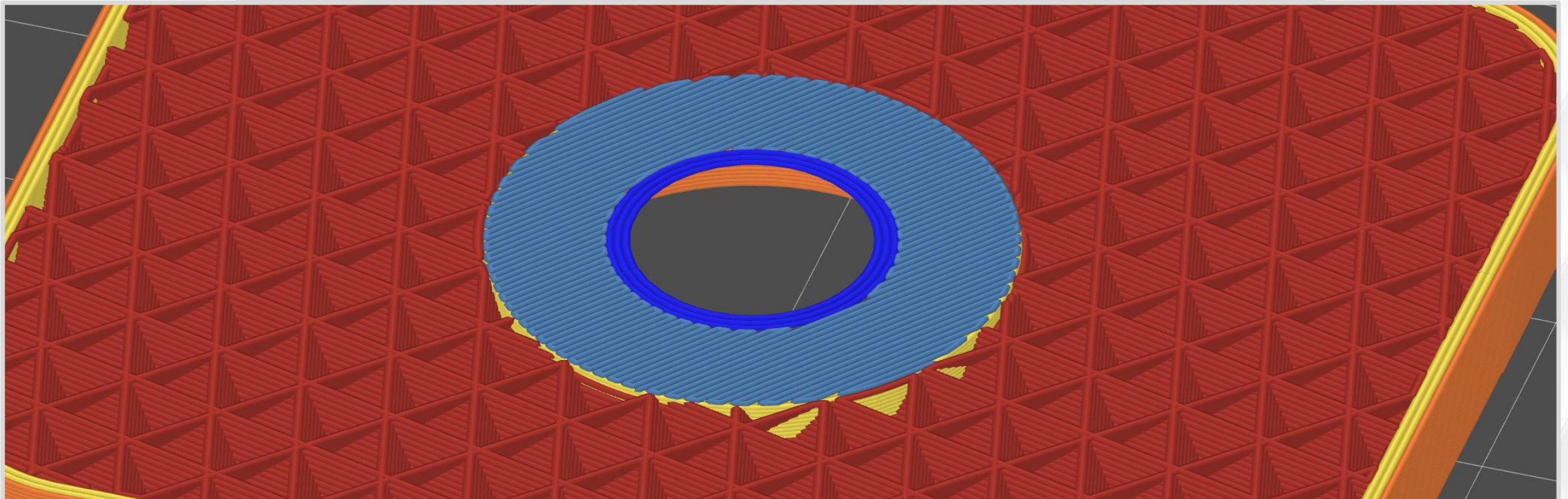


Box Bridge1

Floating bridge anchors, Long bridging extrusions

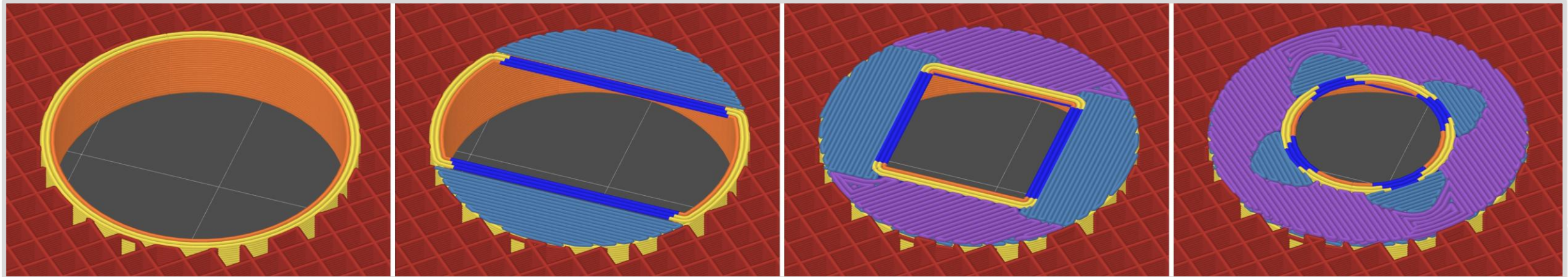
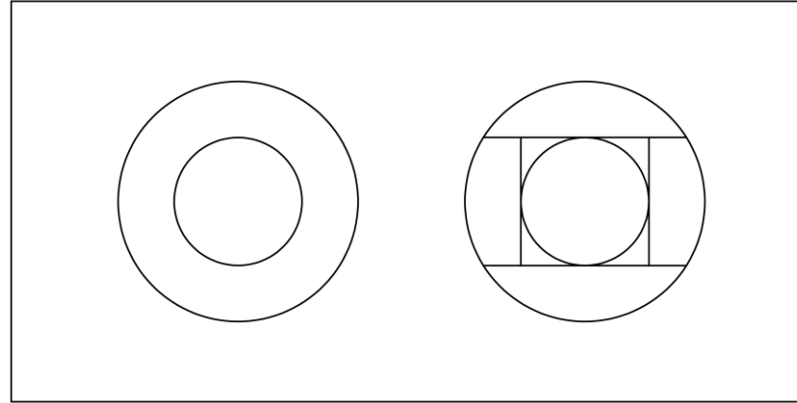


Consider enabling supports.



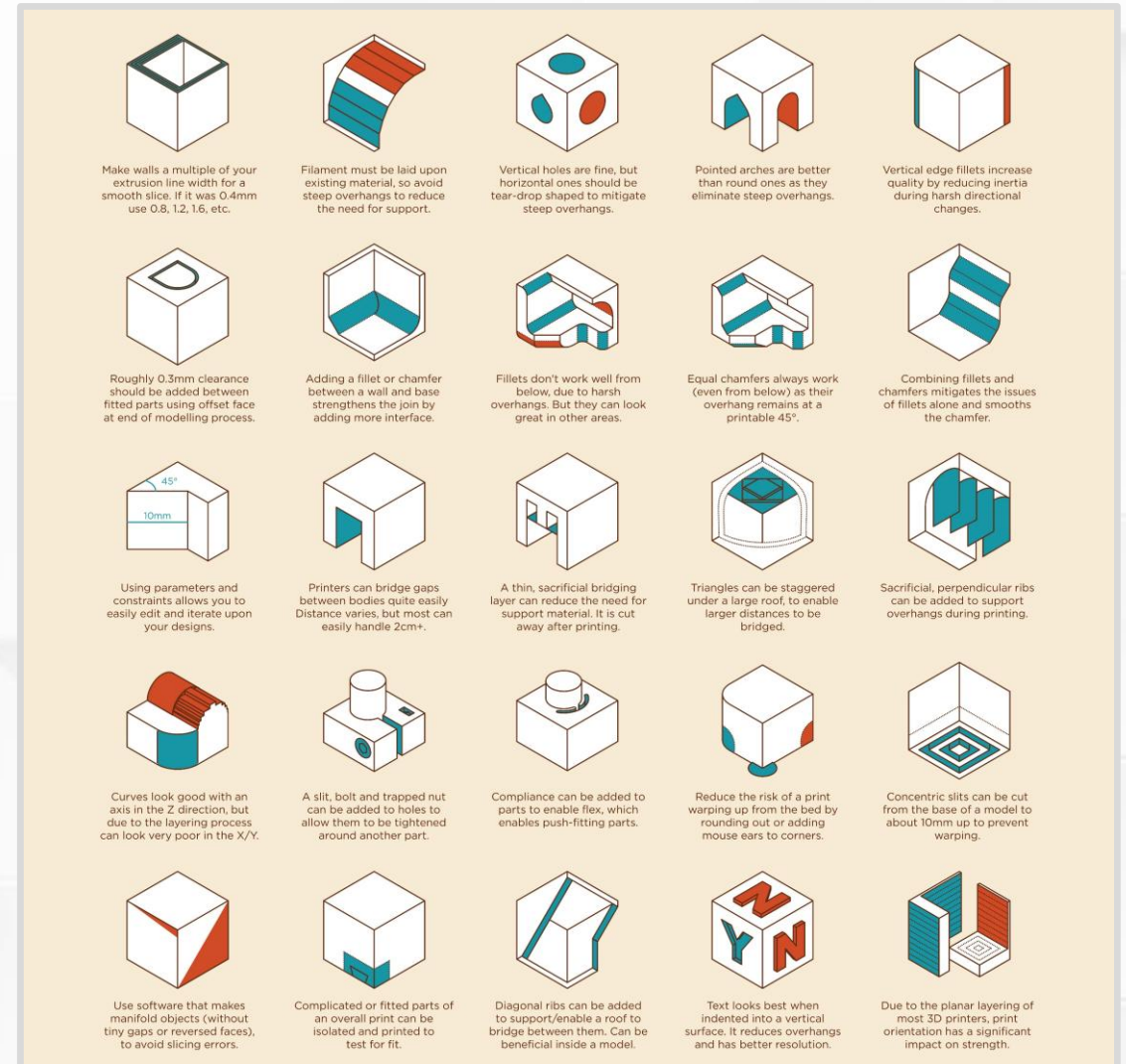
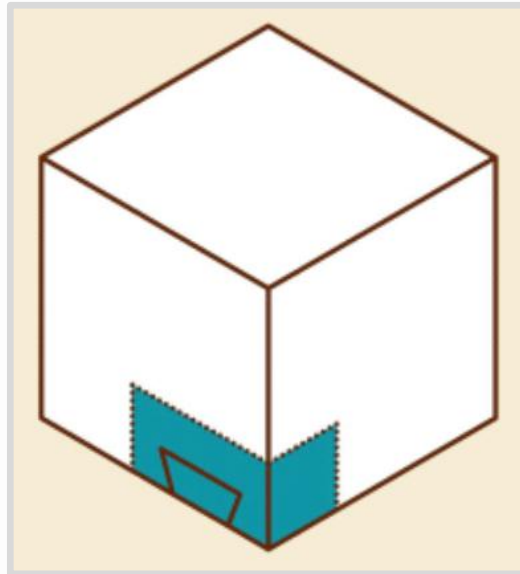
# Sequential Bridges

- There are floating extrusions!
- We can fix this without supports.
- Use right-most drawing to stack three bridges.



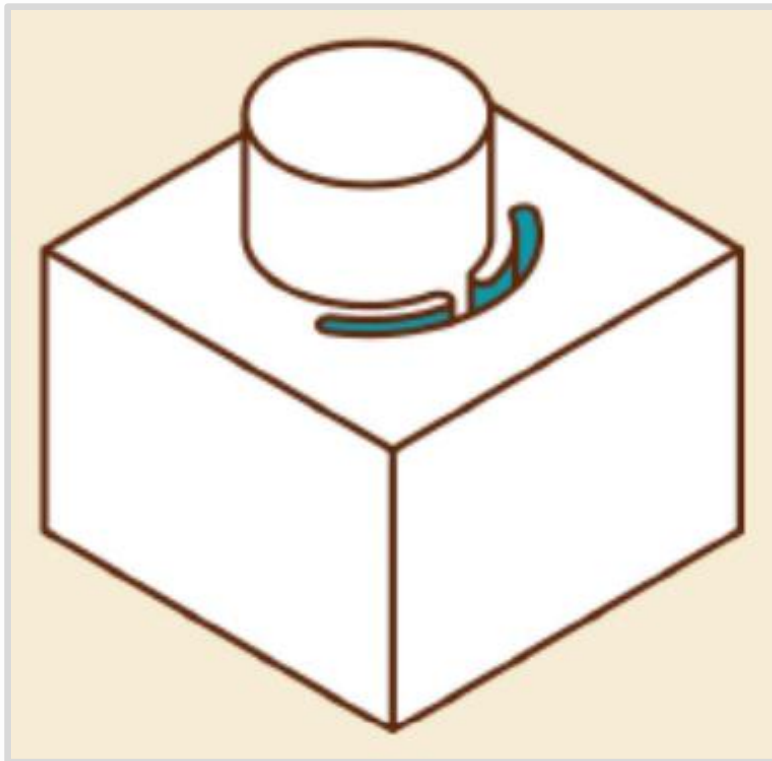
# More Design Tips

- CAD Design Tips for 3D Printing
- Check out this great poster by Billie Ruben.
- “Complicated or fitted parts of a print can be isolated and printed to test for fit.”



# Even More Design Tips

- “Compliance can be added to parts to enable flex, which enables push-fitting parts.”
- “Text looks best when indented into a vertical surface. It reduces overhangs and has a better resolution.”



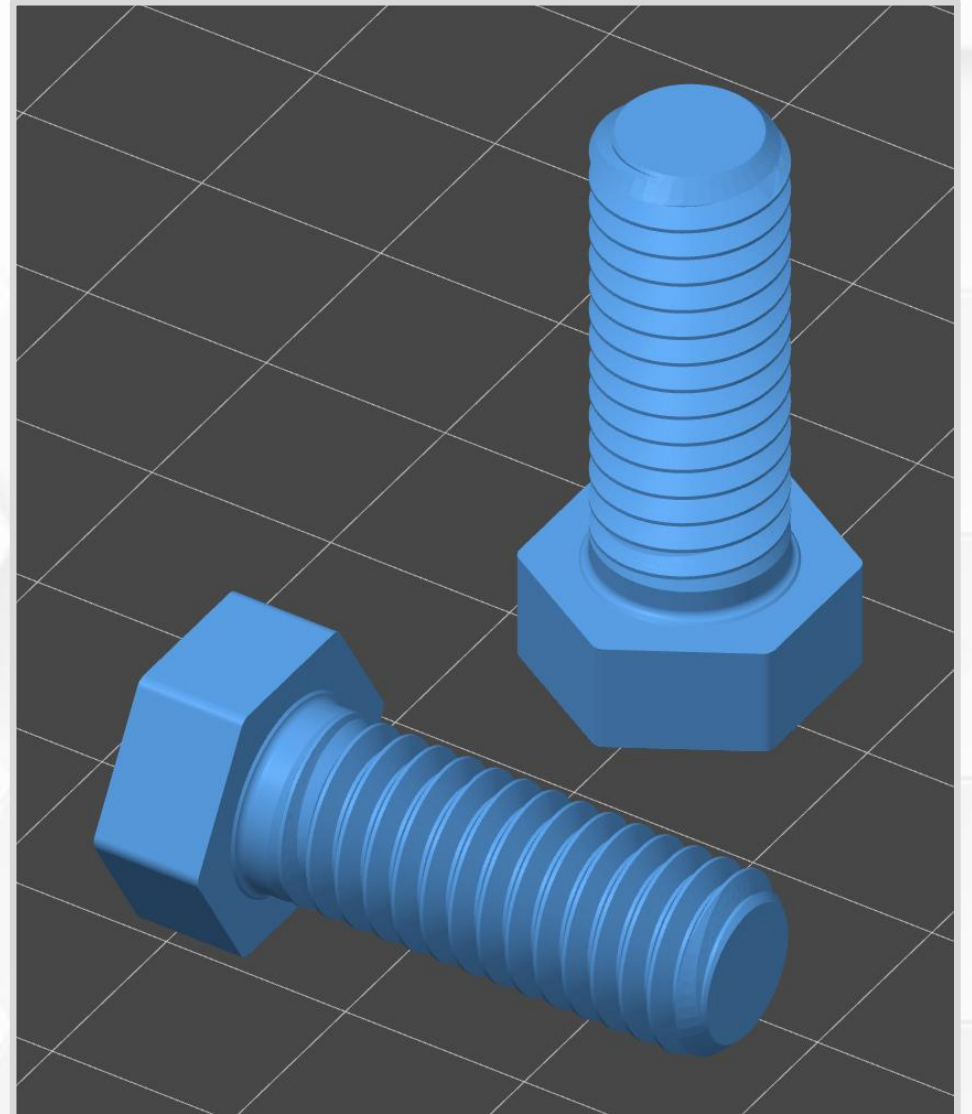
(Tips by  
Billie Ruben)



# Screws

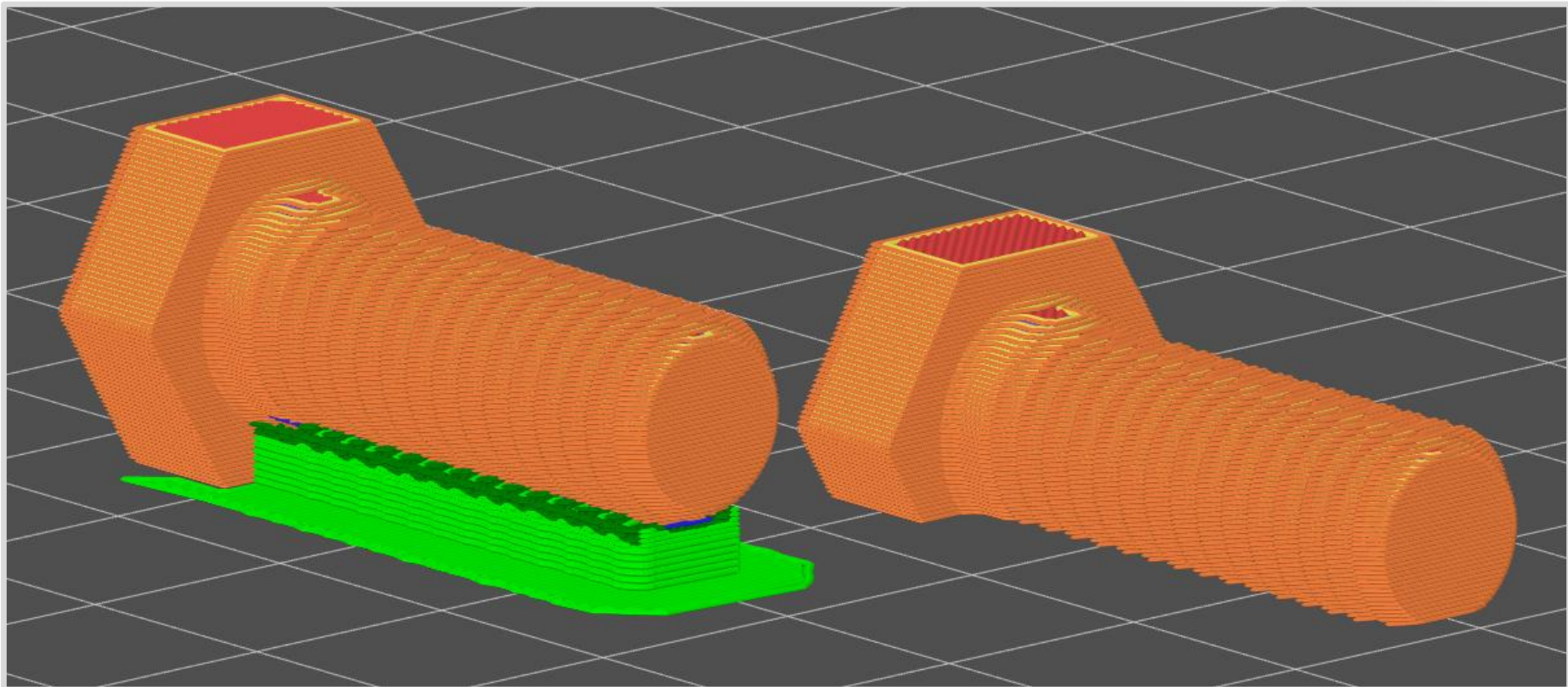
Let's 3D print a plastic screw.

- What differences can you see in strength and printability?
  - Compare and contrast print orientations.
  - Horizontal.
    - Layer lines maximize strength.
    - Nozzle must be smaller than threads.
    - Bottom threads are floating extrusions.
- Suggestions?



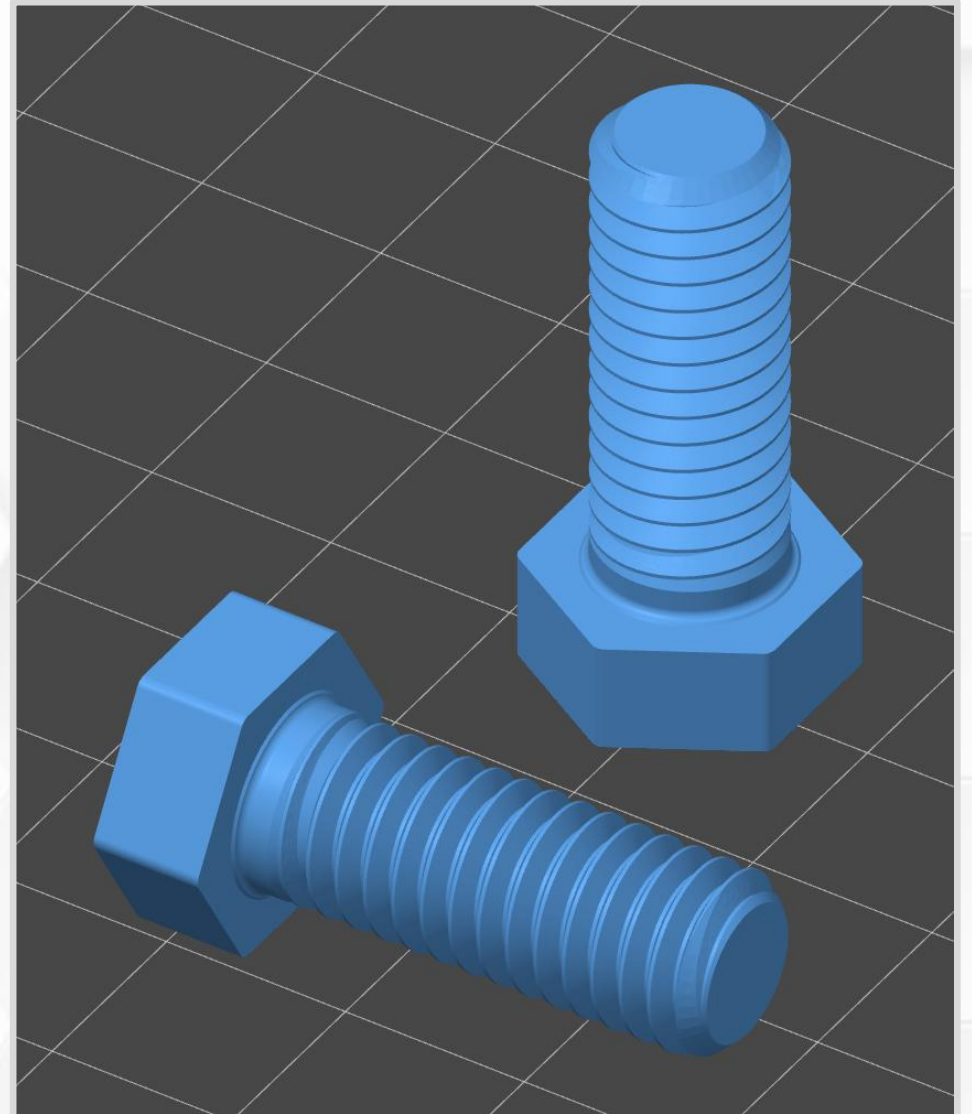
# Screws

- Supports will probably leave a messy surface.
- Trim the bottom so it can sit flat. Clean threads and more bottom surface area.



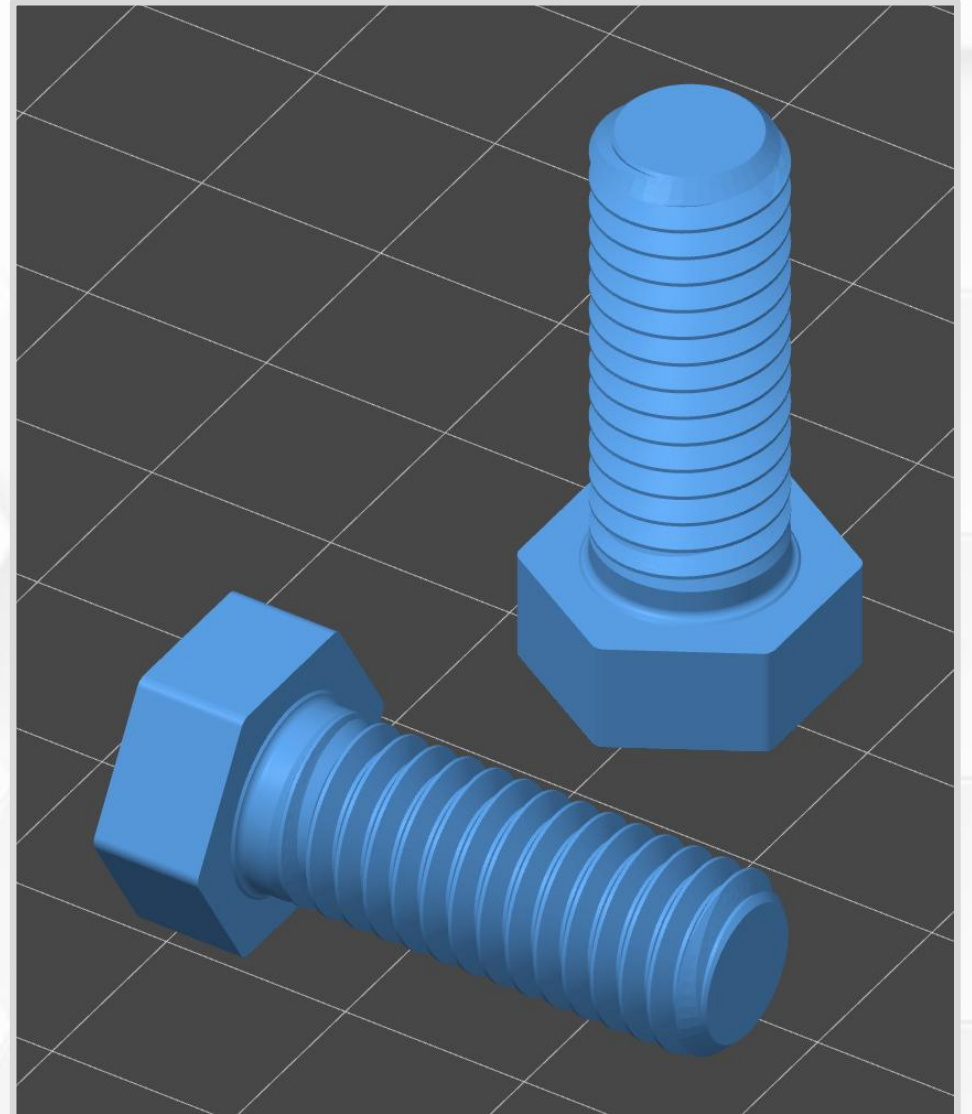
# Screws

- Vertical
  - Thread size limited by layer height.
  - Typical threads create steep overhangs. Compensate with thinner layers.
  - Layers lines create a weaker part.
  - No supports.
  - No modifications required.



# Further Reading

- Custom Threads for 3D printing by [printables.com/@FlyingGyroscope](https://printables.com/@FlyingGyroscope).
  - Parametric template.
  - Includes CAD lesson.
- Recommended viewing.
  - Strength testing printed bolts by My Tech Fun.
  - And a follow-up video.



# Snap Buckle

Let's design a replacement part.

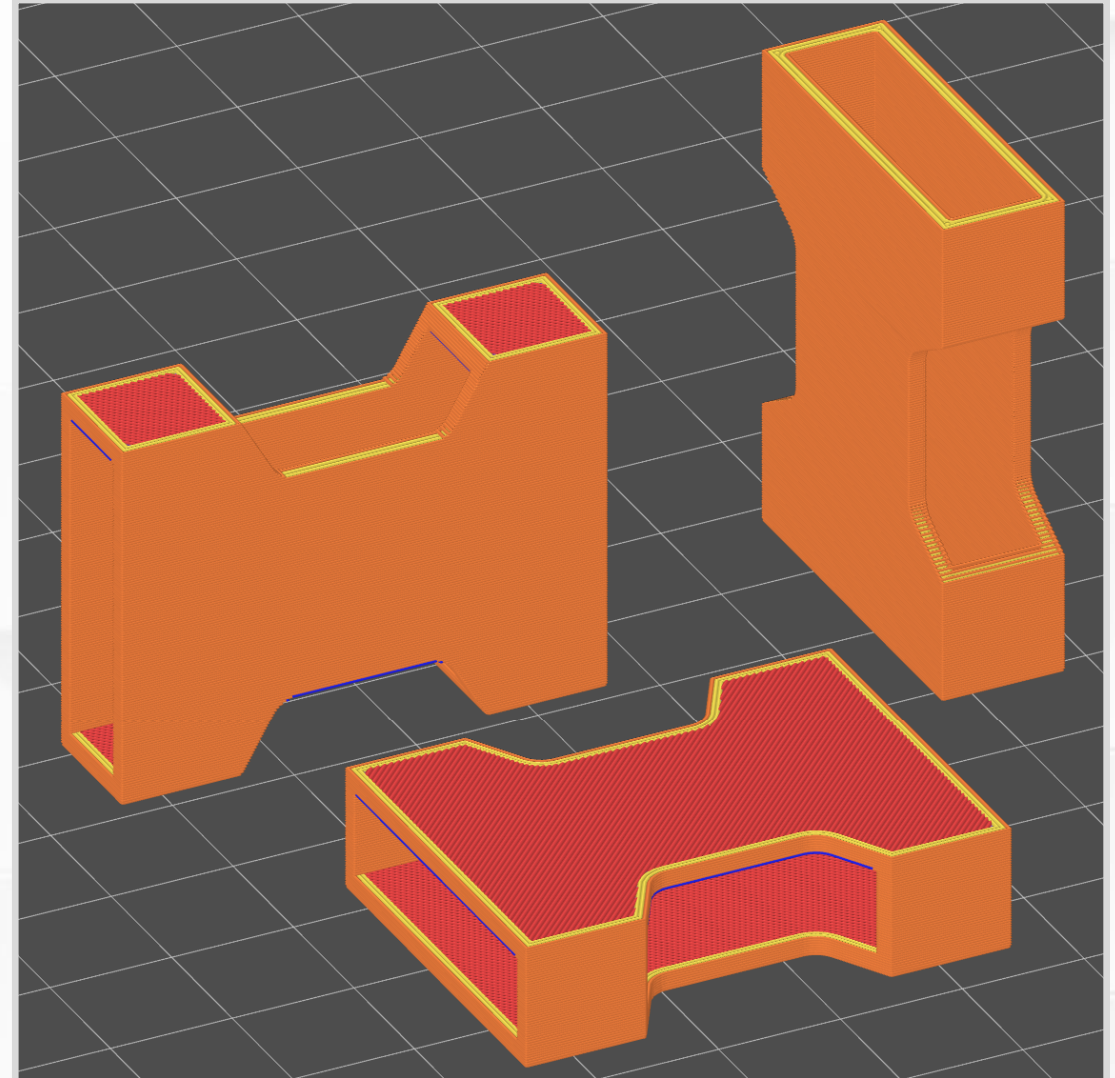
- Should you print a direct copy? Or does it need modifications?
- What 3D printing advantages and disadvantages do you see?



# Snap Buckle

Let's compare several print orientations for the socket.

- How does each one balance **printability** and **part strength**?
- Can you modify the part for high printability and high strength?



# Snap Buckle

- How does this design achieve high strength and high printability?



(Snap buckle by Squinn)

# Snap Buckle

- **Horizontal orientation.**
- High strength for intended use.
- Perfect for the compliant mechanism.
- Good first layer contact.
- **The socket has been reimagined.**
- Dovetail (trapezoid) traps middle prong.
- Overhangs instead of bridges.

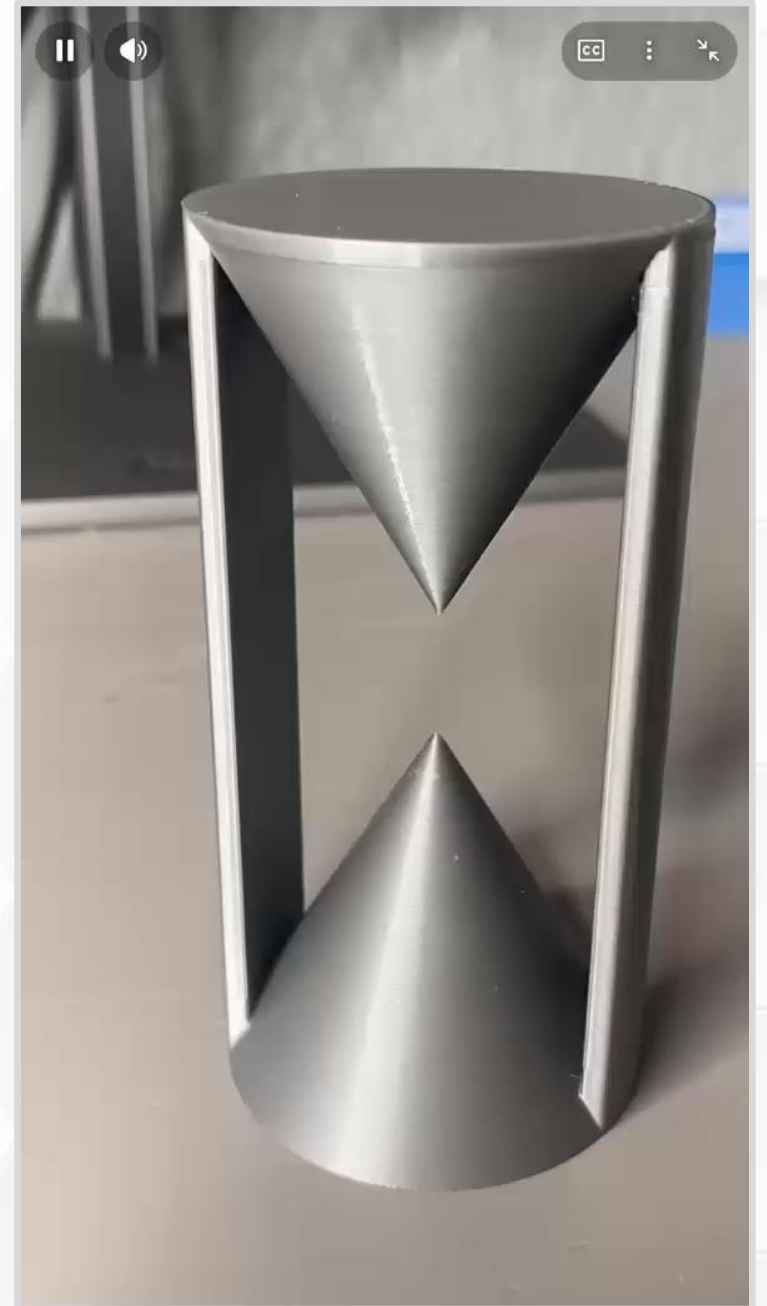
(Snap buckle by Squinn)



# Creative Solutions

- I love the creativity of the maker community.
- This was printed without supports. How?
  - Put a printed part inside a printed part.

(Animation, YouTube short by christopherhelmke)





[github.com/FlyingGyroscope/IntroTo3DPrinting/](https://github.com/FlyingGyroscope/IntroTo3DPrinting/)